One Graph, Four Specialized Execution Plans

Each event type or query compiles to its minimal topologically ordered action sequence.

on(BookUpdate)

1 action · pure state write

`0: Book.handle`

7 downstream nodes skipped; evaluators run lazily when needed

on(Trade)

3 actions · eval + handle

`0: Mid.eval` → `1: Spread.eval` → `2: TradePressure.handle`

Computes only Mid & Spread for TradePressure; skips Imbalance, FairValue, BookBias, Score

on(Reset)

1 action · input-free handler

`0: TradePressure.handle`

on<Reset> declares inputs<>, so even Mid and Spread are skipped

evaluate<Outputs>()

7 actions · shared closure

`0: Mid.eval` → `1: Spread.eval` → `2: Imbalance.eval` → `3: FairValue.eval` → `4: BookBias.eval` → `5: TradePressure.eval` → `6: Score.eval`

Zero handlers run (history preserved); shared Mid, Spread & FairValue each run once -> (score, fair)